



COLF: Disentangled forecasting for non-stationary time series via contrastive learning

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ABSTRACT

Non-stationary time series are widely prevalent in real world, characterized by changing temporal patterns, which greatly challenges the out-of-distribution generalization ability of forecasting approaches. Most existing approaches seek to achieve sequence stationarization through normalization methods, whose effectiveness is still limited since non-stationarity is not simply reflected in shift in mean and variance. Given this, we consider addressing non-stationarity from another perspective by pattern disentanglement, i.e., to distinguish between *stable* and *unstable* patterns according to temporal persistence and utilize them separately for forecasting. Considering the lack of supervised signals for disentanglement as temporal persistence is challenging to measure explicitly, we achieve this in a self-supervised manner and propose a model COLF. Specifically, COLF is designed with two contrastive learning mechanisms at different levels: the *across-sample contrastive learning* to disentangle stable and unstable patterns at sequence level, and the *within-sample contrastive learning* to further extract *persistent components* from unstable patterns under the *local smoothness* assumption at subsequence level. The final forecasted result is obtained by integrating the predictions derived from both stable patterns and the persistent components within unstable patterns. COLF is evaluated on six real-world datasets, and the experimental results demonstrate that COLF achieves overall superior performance compared to state-of-the-art approaches across different forecasting horizons and evaluation metrics.

1. Introduction

Time series forecasting is a widely concerned research problem of great importance to application domains such as economics [1], transportation [2], environmental sciences [3], etc. There are numerous methods for time series forecasting, ranging from statistical time series analysis techniques [4], to the latest machine-learning-based and deep-learning-based approaches [5,6]. Especially, deep-learning-based approaches have exhibited state-of-the-art forecasting performances, and have become a focus of recent research.

Most existing deep-learning-based approaches, although effective in some situations, still do not adapt well to non-stationary time series, which are prevalent in real world. By definition, time series are considered non-stationary if their distributions change over time. Such changes can manifest as variations in statistics such as the mean and covariance or in more complex temporal patterns, such as the evolving

frequency structure of seismic wave signals and the changing momentum effects of asset prices. The key problem raised by non-stationarity is that the *i.i.d.* (independently and identically distributed) assumption no longer holds, and forecasting models may rely on patterns present in the training data but absent at test time, thereby impairing their OOD (out-of-distribution) generalization. To tackle non-stationarity, existing models [7–10] mainly rely on normalization-based stationarization, which essentially seeks to reduce the mean and variance discrepancy across samples. While such stationarization can, to some extent, alleviate the impact of non-stationarity, its effectiveness is still limited since real-world non-stationarity extends beyond shifts in mean and variance.

Due to the limitations of stationarization, we consider tackling non-stationarity through pattern disentanglement. Given a non-stationary time series, we aim to identify which of its patterns persist along the time axis and which are unique to a certain period, where the latter is the factor causing non-stationarity. Here, we refer to these two types

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¹ The data used for plotting are obtained from <https://fred.stlouisfed.org/>.

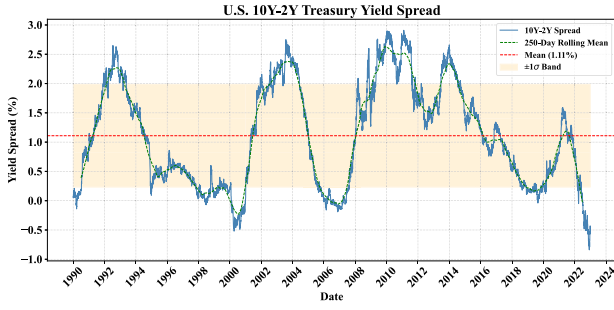


Fig. 1. Yield spread between the 10-year and 2-year U.S. Treasury constant maturity rates from 1990 to 2022. The blue curve represents the daily yield spread, while the dashed green curve shows its 250-day rolling mean. The red dashed line denotes the mean level (1.11%), and the shaded orange area indicates the $\pm 1\sigma$ deviation band.

of patterns as *stable* and *unstable* patterns, respectively. For instance, price spreads in financial markets often display mean-reversion at relatively longer horizons due to the underlying equilibrium relationships or structural constraints, while at shorter horizons, these spreads may experience period-specific deviations driven by market sentiment, liquidity conditions, or event shocks, which in turn introduces non-stationarity into their dynamics, as illustrated in Fig. 1.¹ Since each time series sample can be reliably extrapolated based on stable patterns due to their temporal consistency, the main challenge of forecasting lies in how to properly tackle the unstable patterns. On one hand, fully overlooking unstable patterns may lead to the incompleteness of forecasting; on the other hand, directly leveraging them without control may introduce bias as they may vanish within forecasting horizon. In light of the pervasive *local smoothness* of physical systems, where the high-order differentiability of the systems' state functions ensures their local predictability, represented by heat conduction, fluid dynamics, communication signal generation, etc., we assume that within the unstable patterns, there still exist some components exhibiting local smoothness, which we refer to as *persistent components*, to persist within the forecasting horizon. Extracting these persistent components for forecasting as a supplement to the stable pattern extrapolation is expected to significantly enhance the forecasting performance.

In addition, unlike the classic seasonal-trend decomposition of time series, there lacks explicit information to provide supervision for either pattern disentanglement or persistent component extraction. To address this issue, we argue that a promising method is to use contrastive learning to identify patterns in a self-supervised manner, with a schematic diagram illustrated in Fig. 2. Specifically, an *across-sample contrastive learning* mechanism is proposed to distinguish temporally shared stable patterns from sample-specific unstable patterns, while a *within-sample contrastive learning* mechanism is further introduced to extract components that remain locally persistent across subsequences within each sample.

Based on the discussions above, in this paper, we propose a novel model, named **C**ontrastive- **L**earning-based time series **F**orecaster (**COLF**) for non-stationary time series forecasting. COLF comprises four modules: a disentangling module, two forecasting modules, and an integration module. The disentangling module adopts a Transformer-Encoder architecture, specifically equipped with a disentangling attention mechanism and a sequence-wise across-sample contrastive learning mechanism, to derive stable and unstable representations from each sample's past sequence. The stable representation is directly extrapolated to form the forecasting backbone, whereas persistent components are further extracted from the unstable representation through subsequence-wise within-sample contrastive learning and then used to generate a complementary forecast that augments the backbone. Both forecasting modules adopt an encoder-decoder architecture consisting

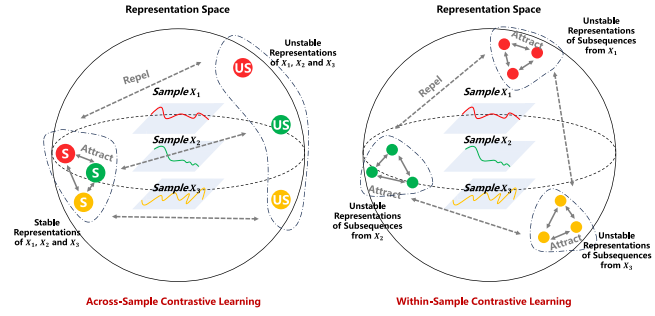


Fig. 2. Illustration of how the proposed *across-sample contrastive learning* and *within-sample contrastive learning* work to achieve pattern disentanglement and persistent component extraction.

of a variate-token Transformer Encoder and a linear decoder. Finally, the integration module combines the two forecasts to produce the final prediction. The main contributions are summarized as follows.

- A novel model COLF is proposed for non-stationary time series forecasting, which tackles non-stationarity through contrastive-learning-based pattern disentanglement.
- Two contrastive learning mechanisms are designed at different levels: an across-sample contrastive learning mechanism to facilitate the disentanglement of stable and unstable patterns at the sequence level, and a within-sample contrastive learning mechanism to further extract persistent components from unstable patterns at the subsequence level under the local smoothness assumption.
- Theoretical analyses are conducted to justify the proposed framework, and extensive experiments on real-world datasets are performed to demonstrate its overall superior forecasting performance over state-of-the-art approaches and its effectiveness in pattern disentanglement.

The rest of this article is organized as follows. In Section 2, we review the related work of this article. In Section 3, we formally introduce the proposed method. In Section 4, we present and discuss the experimental results in detail. And the final conclusions are provided in Section 5.

2. Related work

2.1. Time series forecasting

The statistical time series analysis approaches are the most classic group of approaches in the field of time series forecasting, including ARMA-family models [11] mainly characterizing conditional mean dynamics, GARCH-family models [12] for time-varying conditional variance modeling, nonlinear models such as TAR [13] and MSA [14], and even composite methods that integrate forecasts from multiple statistical specifications to enhance robustness as shown in [15,16]. These approaches are simple in form, highly interpretable, and widely used in practice. To better capture the nonlinearity and complex patterns in some time series data, an increasing number of studies have begun to apply machine learning to time series forecasting. Among them, some are based on traditional machine learning, such as SVM-based [17,18] and decision-tree-based [19,20] approaches. In recent years, however, more works have turned to deep learning to exploit its powerful end-to-end sequence modeling capabilities for improved forecasting performance. According to their primary architectural designs, deep-learning-based approaches can be broadly categorized into several families, including CNN-based approaches, such as TCN [21] and MICN [22], which model temporal patterns through convolutional filters; RNN-based approaches, such as DeepAR [23] and ATFN [24], which are

mainly based on recurrent neural network variants LSTM and GRU in pursuit of more natural modeling of temporal patterns; Transformer-based approaches which rely on Transformer architectures and attention mechanisms to extract temporal features and model temporal correlations, represented by FEDformer [25], PatchTST [26], Non-stationary Transformer [9] and Koopa [27], with the latter two specifically designed to address non-stationarity; GNN-based approaches such as GTS [28] and PMEDGN [29], which are characterized by leveraging spatial features through graph neural networks to improve multivariate time series forecasting performance; NDE-based approaches such as ETN-ODE [30] and MODE [31], which employ neural differential equations to model temporal dynamics in a continuous way; and MLP-based approaches such as DLinear [32], TSMixer [33] and Patch-FusionMLP [34], which adopt simple linear or MLP-based architectures for time series modeling and are favored for their simplicity and computational efficiency.

2.2. Contrastive learning

Contrastive learning is a type of self-supervised learning to learn representations from unlabeled data, built on the idea that effective representation learning can be achieved by making similar samples (positive samples) close to each other while pushing dissimilar samples (negative samples) apart in representation space in the absence of labels. Typical contrastive learning approaches include CPC [35], which learns representations by predicting future latent variables, MoCo [36], which introduces a momentum encoder and a dynamic dictionary for efficient contrastive learning, and SimCLR [37], which simplifies contrastive learning with a minimalist design based on data augmentation and large-batch training. All of these are universal frameworks that can be applied to many domains such as computer vision, natural language processing, time series analysis, speech processing and so on. Also, there are several works tailored to perform contrastive learning for time series data, such as CoST [38], which aims to learn disentangled seasonal-trend representations via contrastive learning, CLUDA [39], which is designed specifically for unsupervised domain adaptation of time series data, TS2Vec [40], which is featured by the designed hierarchical contrasting and contextual consistency mechanisms, MF-CLR [41], which proposes a contrastive learning framework to learn representations of time series across multiple frequencies, etc.

3. Methodology

In this section, we formally introduce COLF. Given a set of D -variate non-stationary time series samples $S = \{S_i | S_i = (X_i, Y_i), i = 1, 2, \dots, N\}$ collected along the time axis, where $X_i \in \mathbb{R}^{D \times T}$ and $Y_i \in \mathbb{R}^{D \times L}$ denote the T -length past sequence and L -length future sequence of i th sample S_i , respectively, COLF aims to find a map $f : \mathbb{R}^{D \times T} \rightarrow \mathbb{R}^{D \times L}$ such that forecasted sequence $\hat{Y}_i = f(X_i)$ can serve as an effective estimate of Y_i . The biggest challenge here is posed by non-stationarity, requiring forecasting models to have effective OOD generalization capabilities.

3.1. Overall framework

3.1.1. Challenge of non-stationary time series forecasting - A causal view

We first offer an explanation of *why non-stationarity poses a great challenge for forecasting*, from the perspective of causality.

Following the notations above, the temporal patterns of any given X_i can be distinguished into two categories: the *stable patterns* that are shared by all sequences in $\{X_i | i = 1, 2, \dots, N\}$, and the *unstable patterns* that are unique to the current X_i . The unstable patterns are the source of the OOD problem since they are period-specific and may not persist into the future. However, most existing time series forecasting models do not explicitly distinguish between these patterns. As illustrated by the structural causal model [42] in Fig. 3, they derive the representation Z indiscriminately from both stable patterns P^s and

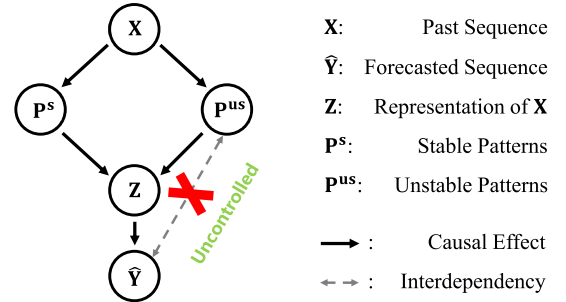


Fig. 3. Time series forecasting without adjustments for non-stationarity.

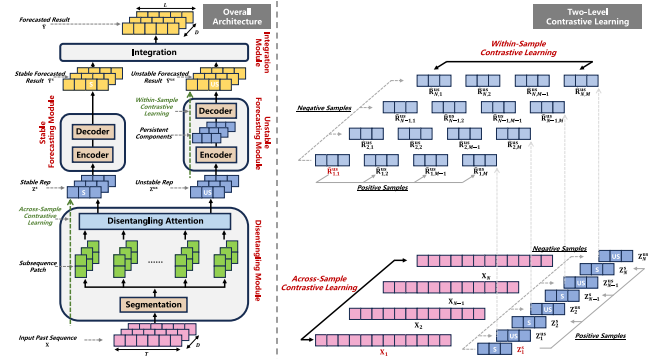


Fig. 4. Overall architecture of the proposed COLF. The left subfigure exhibits the architecture of COLF (four modules, two contrastive learning mechanisms) and the overall workflows, while the right subfigure provides a diagram of the designed contrastive learning mechanisms.

unstable patterns P^{us} , and then use it to generate the forecast $\hat{Y}$. Even though there is no direct causal effect between P^{us} and $\hat{Y}$, the existing *backdoor path*, i.e., $P^{us} \leftarrow X \rightarrow P^s \rightarrow Z \rightarrow \hat{Y}$, makes P^{us} and $\hat{Y}$ interdependent. When unstable patterns change or disappear in the testing data, which often happens since they lack temporal persistence, forecasting models may encounter a sharp drop in performance if they still pay much attention to unstable patterns without control.

Therefore, to achieve effective non-stationary time series forecasting, it is necessary to distinguish between P^s and P^{us} , and treat them differently. Besides, the use of P^{us} in forecasting also needs to be conducted under some level of control to mitigate the negative effects that P^{us} may introduce due to their temporal non-persistence.

3.1.2. Model architecture

To address the challenge discussed above, we propose COLF to perform non-stationary time series forecasting through a workflow including *stable-unstable pattern disentanglement*, *pattern-wise forecasting* and *forecasted result integration*, implemented by four internal modules, i.e., a disentangling module $\mathcal{M}_{\text{disent}}$, a forecasting module for stable patterns $\mathcal{M}_{\text{fctst}}^s$, a forecasting module for unstable patterns $\mathcal{M}_{\text{fctst}}^{us}$, and an integration module $\mathcal{M}_{\text{int}}$. For any given input sequence X_i , its future sequence is predicted through:

$$\begin{aligned} Z_i^s, Z_i^{us} &= \mathcal{M}_{\text{disent}}(X_i), \quad \hat{Y}_i^s = \mathcal{M}_{\text{fctst}}^s(Z_i^s), \quad \hat{Y}_i^{us} = \mathcal{M}_{\text{fctst}}^{us}(Z_i^{us}), \\ \hat{Y}_i &= \mathcal{M}_{\text{int}}(\hat{Y}_i^s, \hat{Y}_i^{us}), \end{aligned} \quad (1)$$

where Z_i^s and Z_i^{us} denote the disentangled stable representation and unstable representation, $\hat{Y}_i^s$ and $\hat{Y}_i^{us}$ denote the forecasted results based on Z_i^s and Z_i^{us} , and $\hat{Y}_i$ denotes the final forecasted result. The entire architecture of COLF is exhibited in Fig. 4. The details of $\mathcal{M}_{\text{disent}}$, $\mathcal{M}_{\text{fctst}}^s$ and $\mathcal{M}_{\text{fctst}}^{us}$ will be discussed in the following subsections. And for $\mathcal{M}_{\text{int}}$, we specify it as a simple equal-weight additive integration, which is enough to achieve good performance.

3.2. Pattern disentanglement

3.2.1. Disentangling attention

The disentangling attention mechanism involves two negatively correlated attention mechanisms to extract contrasting temporal patterns from the same input sequence:

$$F_{\text{attn}}^s(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{H}}\right)\mathbf{V}, \quad F_{\text{attn}}^u(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(-\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{H}}\right)\mathbf{V}, \quad (2)$$

where $F_{\text{attn}}^s(\cdot)$ and $F_{\text{attn}}^u(\cdot)$ denote the pair of contrasting attention functions, $\mathbf{Q}$, $\mathbf{K}$ and $\mathbf{V}$ denote the corresponding query, key and value matrices, and H denotes the dimension of representation space. It is worth noting that $F_{\text{attn}}^s(\cdot)$ and $F_{\text{attn}}^u(\cdot)$ are designed to construct two complementary representation streams rather than to directly guarantee stable-unstable semantic separation. The semantic distinction between the two streams is further induced by the across-sample contrastive learning mechanism introduced in Section 3.2.2. Also, different from the pair of negatively correlated attention mechanisms used in [43], which are applied at the point level, our disentangling attention defined here is applied at the subsequence level, since modeling temporal correlations at the subsequence level is considered to be more effective than point level for time series data as discussed in [26,44,45].

Specifically, the disentangling attention mechanism functions in a variate-independent way, as suggested by [26]. For each univariate sequence $\mathbf{x}_d = (x_{d,1}, x_{d,2}, \dots, x_{d,T}) \in \mathbb{R}^T$ from input $\mathbf{X} \in \mathbb{R}^{D \times T}$, $\mathbf{x}_d$ is first segmented into a patch of subsequences $\mathcal{P}_d = \{x_{d,m:m+l} \in \mathbb{R}^l | m = 1, 2, \dots, M\}$ according to subsequence length l and stride s . In practice, we also pad $\mathbf{x}_d$ at the end with its last value x_T for s times before segmenting, leading to $M = \lfloor \frac{T-l}{s} \rfloor + 2$. With $\mathcal{P}_d$ packed into matrix $\mathbf{P}_d \in \mathbb{R}^{M \times l}$, which is further projected into $\mathbf{Q}_d \in \mathbb{R}^{M \times H}$, $\mathbf{K}_d \in \mathbb{R}^{M \times H}$ and $\mathbf{V}_d \in \mathbb{R}^{M \times H}$, $\mathbf{x}_d$'s disentangled representations $\mathbf{z}_d^s \in \mathbb{R}^{M \times H}$ and $\mathbf{z}_d^u \in \mathbb{R}^{M \times H}$ can be computed by:

$$\mathbf{z}_d^s = F_{\text{enc}} \circ F_{\text{attn}}^s(\mathbf{Q}_d, \mathbf{K}_d, \mathbf{V}_d), \quad \mathbf{z}_d^u = F_{\text{enc}} \circ F_{\text{attn}}^u(\mathbf{Q}_d, \mathbf{K}_d, \mathbf{V}_d), \quad (3)$$

where $F_{\text{enc}}: \mathbb{R}^{M \times H} \rightarrow \mathbb{R}^{M \times H}$ involves all other computations following the paradigm of vanilla Transformer Encoder, and $\circ$ denotes function composition. Based on this, the disentangled stable and unstable representations of $\mathbf{X}$, $\mathbf{Z}^s \in \mathbb{R}^{D \times M \times H}$ and $\mathbf{Z}^u \in \mathbb{R}^{D \times M \times H}$, can be obtained by concatenating disentangled representations of all $\mathbf{x}_d$:

$$\mathbf{Z}^s = [\mathbf{z}_1^s, \mathbf{z}_2^s, \dots, \mathbf{z}_D^s], \quad \mathbf{Z}^u = [\mathbf{z}_1^u, \mathbf{z}_2^u, \dots, \mathbf{z}_D^u]. \quad (4)$$

3.2.2. Across-sample contrastive learning

Recall that the stable and unstable representations obtained solely by disentangling attention mechanism are only negatively correlated, and do not yet reflect the stable and unstable patterns of input past sequence. Therefore, we further introduce an across-sample contrastive learning mechanism to help achieve this.

For all input samples $\{\mathbf{X}_i \in \mathbb{R}^{D \times T} | i = 1, 2, \dots, N\}$, let $\{\mathbf{Z}_i^s \in \mathbb{R}^{D \times M \times H} | i = 1, 2, \dots, N\}$ and $\{\mathbf{Z}_i^u \in \mathbb{R}^{D \times M \times H} | i = 1, 2, \dots, N\}$ denote the stable and unstable representations obtained by disentangling attention, respectively. Since stable patterns refer to patterns shared by all samples, stable representations of all samples should be similar with each other. In contrast, due to the temporal uniqueness of unstable patterns, each sample's unstable representation should differ from all stable representations, including both its own stable representation and those of other samples. From the view of contrastive learning, this means that all representations in $\{\mathbf{Z}_i^s \in \mathbb{R}^{D \times M \times H} | i = 1, 2, \dots, N\}$ form positive samples to each other, while for each $\mathbf{Z}_i^s$, all representations in $\{\mathbf{Z}_i^u \in \mathbb{R}^{D \times M \times H} | i = 1, 2, \dots, N\}$ form its negative samples. As such, we

define a contrastive loss $\mathcal{L}_{\text{across}}$ as follows:

$$\mathcal{L}_{\text{across}} = \frac{-1}{DN(N-1)} \sum_{d=1}^D \sum_{i=1}^N \sum_{\substack{i^+=1 \\ i^+ \neq i}}^N \log \frac{\exp(\psi_d(\mathbf{Z}_i^s) \cdot \psi_d(\mathbf{Z}_{i^+}^s)/\tau)}{\exp(\psi_d(\mathbf{Z}_i^s) \cdot \psi_d(\mathbf{Z}_{i^+}^s)/\tau) + \sum_{j=1}^N \exp(\psi_d(\mathbf{Z}_i^s) \cdot \psi_d(\mathbf{Z}_j^u)/\tau)}, \quad (5)$$

where $\psi_d: \mathbb{R}^{D \times M \times H} \rightarrow \mathbb{R}^{MH}$ is a function to first extract d th element along the variate dimension and then flatten the last two dimensions, representations with indices i and i^+ form a positive pair, and τ is the temperature parameter, which controls the sharpness of the contrastive distribution and the discrimination between positive and negative representations. In practice, N is usually set to the batch size, which can be very large on some occasions. To reduce the amount of computation, we can randomly select K positive samples for computation and thus use a simplified version of $\mathcal{L}_{\text{across}}$ defined in Eq. (6):

$$\mathcal{L}_{\text{across}}^{\text{sim}} = \frac{-1}{DNK} \sum_{d=1}^D \sum_{i=1}^N \sum_{k=1}^K \log \frac{\exp(\psi_d(\mathbf{Z}_i^s) \cdot \psi_d(\mathbf{Z}_{i_k^+}^s)/\tau)}{\exp(\psi_d(\mathbf{Z}_i^s) \cdot \psi_d(\mathbf{Z}_{i_k^+}^s)/\tau) + \sum_{j=1}^N \exp(\psi_d(\mathbf{Z}_i^s) \cdot \psi_d(\mathbf{Z}_j^u)/\tau)}, \quad (6)$$

where $i_k^+ \sim \mathcal{U}(\{1, 2, \dots, N\} \setminus \{i\})$ follows a uniform distribution over $\{1, 2, \dots, N\} \setminus \{i\}$, and the representation with index i_k^+ serves as a positive sample for that with index i . Let $\mathbf{I}_K^+$ denote the collection of indices of the positive samples independently and uniformly drawn for all i , and let θ denote the trainable model parameters. Conditional on the current batch $\mathcal{B}$, $\mathcal{L}_{\text{across}}^{\text{sim}}$ and $\nabla_{\theta} \mathcal{L}_{\text{across}}^{\text{sim}}$ are unbiased estimators of $\mathcal{L}_{\text{across}}$ and $\nabla_{\theta} \mathcal{L}_{\text{across}}$, respectively:

$$\mathbb{E}_{\mathbf{I}_K^+} [\mathcal{L}_{\text{across}}^{\text{sim}} | \mathcal{B}] = \mathcal{L}_{\text{across}}, \quad \mathbb{E}_{\mathbf{I}_K^+} [\nabla_{\theta} \mathcal{L}_{\text{across}}^{\text{sim}} | \mathcal{B}] = \nabla_{\theta} \mathcal{L}_{\text{across}}. \quad (7)$$

Thus, this approximation introduces no systematic bias, while its sampling-induced fluctuations also decrease as K increases. The proof is provided in Section 2 of the supplementary material.

3.2.3. Theoretical justification

We further provide theoretical insights into how the proposed across-sample contrastive learning mechanism promotes the separation of stable and unstable patterns.

Note that the across-sample contrastive loss defined in Eq. (5) is computed in a variate-wise manner. For the d th variate, the loss can be written as

$$\mathcal{L}_{\text{across}}^d = \mathbb{E}_{(S, S^+), U^{\text{anc}}, U^{\text{pos}}, U_{1:N-2}} \left[-\log \frac{f(S, S^+)}{f(S, S^+) + f(S, U^{\text{anc}}) + f(S, U^{\text{pos}}) + \sum_{j=1}^{N-2} f(S, U_j)} \right], \quad (8)$$

where $S \in \mathbb{R}^{MH}$ is the anchor, $S^+ \in \mathbb{R}^{MH}$ is the corresponding positive sample, and $U^{\text{anc}} \in \mathbb{R}^{MH}$, $U^{\text{pos}} \in \mathbb{R}^{MH}$ and $U_{1:N-2} = (U_1, U_2, \dots, U_{N-2})$ together form the negative samples, with $U_j \in \mathbb{R}^{MH}$ for $j = 1, 2, \dots, N-2$. Here, $f(\cdot, \cdot)$ is defined by $f(x, y) = \exp(x \cdot y/\tau)$. Specifically, U^{anc} and U^{pos} are unstable representations derived from the same input samples as S and S^+ , respectively, while $U_{1:N-2}$ are unstable representations derived from the remaining other samples.

Proposition 1. Assume that the positive pair (S, S^+) is sampled from $P_{(S, S^+)}$, and that the negatives $U_1, U_2, \dots, U_{N-2}$ are independently sampled from P_U and are independent of (S, S^+) . If $P_{S^+|S=S} \ll P_U$ for P_S -almost every s , then

$$D_{\text{KL}}(P_{(S, S^+)} \parallel P_S \otimes P_U) \geq \log(N-1) - \mathcal{L}_{\text{ind}}^d, \quad (9)$$

where

$$\mathcal{L}_{\text{ind}}^d = \mathbb{E}_{(S, S^+), U_{1:N-2}} \left[-\log \frac{f(S, S^+)}{f(S, S^+) + \sum_{j=1}^{N-2} f(S, U_j)} \right]. \quad (10)$$

Moreover,

$$D_{\text{KL}}(P_{(S,S^+)} \parallel P_S \otimes P_U) = I(S; S^+) + D_{\text{KL}}(P_{S^+} \parallel P_U). \quad (11)$$

Since $\mathcal{L}_{\text{across}}^d$ additionally includes U^{anc} and U^{pos} , we have $\mathcal{L}_{\text{across}}^d \geq \mathcal{L}_{\text{ind}}^d$. By Proposition 1, minimizing $\mathcal{L}_{\text{across}}^d$ raises a valid, although more conservative, lower bound on the sum of the mutual information $I(S; S^+)$ and KL divergence $D_{\text{KL}}(P_{S^+} \parallel P_U)$, thereby encouraging stable pattern consistency and stable-unstable pattern separation. In addition, although U^{anc} and U^{pos} are not independent of the current positive pair (S, S^+) and are therefore not directly covered by the proposition, they are indirectly encouraged to move away from S^+ during training, as they can serve as independent negatives for other positive pairs. The proof and relevant notation are provided in Section 1 of the supplementary material.

3.3. Pattern-wise forecasting

3.3.1. Forecasting architecture

Technically, both forecasting modules, $\mathcal{M}_{\text{fctst}}^s$ and $\mathcal{M}_{\text{fctst}}^{\text{us}}$, share the same forecasting architecture, specifically an encoder-decoder framework. For simplicity, we use $\mathbf{Z} \in \mathbb{R}^{D \times M \times H}$ to denote any representation of input $\mathbf{X} \in \mathbb{R}^{D \times T}$, whether stable or unstable. The forecasted sequence $\hat{\mathbf{Y}} \in \mathbb{R}^{D \times L}$ is generated through:

$$\tilde{\mathbf{Z}} = f_{\text{proj}}(\mathbf{Z}), \quad \hat{\mathbf{Z}} = F_{\text{enc}}(\text{softmax}(\frac{\tilde{\mathbf{Q}}\tilde{\mathbf{K}}^T}{\sqrt{H}})\tilde{\mathbf{V}}), \quad \hat{\mathbf{Y}} = F_{\text{gen}}(\hat{\mathbf{Z}}). \quad (12)$$

Here, $\tilde{\mathbf{Z}} \in \mathbb{R}^{D \times H}$ is the lower dimensional projection of $\mathbf{Z}$, with $\tilde{\mathbf{Q}} \in \mathbb{R}^{D \times H}$, $\tilde{\mathbf{K}} \in \mathbb{R}^{D \times H}$ and $\tilde{\mathbf{V}} \in \mathbb{R}^{D \times H}$ being its corresponding query, key and value, and $\hat{\mathbf{Z}} \in \mathbb{R}^{D \times H}$ is the encoded result of $\tilde{\mathbf{Z}}$. The projector $f_{\text{proj}}: \mathbb{R}^{D \times M \times H} \rightarrow \mathbb{R}^{D \times H}$ is parameterized as a stack of a flatten layer to flatten the last two dimensions and a linear layer for dimension reduction. F_{enc} is the same as that in Eq. (3), and $F_{\text{gen}}: \mathbb{R}^{D \times H} \rightarrow \mathbb{R}^{D \times L}$ is parameterized as a linear layer. Specifically, the attention mechanism applied here functions based on variate token, as suggested by [45], to further capture temporal patterns, as well as to model correlations between multiple variates.

With the same forecasting architecture, the difference between $\mathcal{M}_{\text{fctst}}^s$ and $\mathcal{M}_{\text{fctst}}^{\text{us}}$ lies in what features are expected to be learned from the input representations in preparation for subsequent decoding. As stable patterns can be directly extrapolated due to the temporal consistency, the encoder in $\mathcal{M}_{\text{fctst}}^s$ is directly applied to stable representation $\mathbf{Z}^s$ without any control over feature learning. In contrast, $\mathcal{M}_{\text{fctst}}^{\text{us}}$ aims to extract *persistent components* from unstable representation $\mathbf{Z}^{\text{us}}$, and leverage them to perform forecasting, which thus imposes requirements on the feature learning in $\mathcal{M}_{\text{fctst}}^{\text{us}}$. To facilitate this, we introduce a within-sample contrastive learning mechanism for $\mathcal{M}_{\text{fctst}}^{\text{us}}$.

3.3.2. Within-sample contrastive learning

Recall that persistent components refer to those parts existing in unstable patterns, which may still persist within the forecasting horizon. The key to discover these components is to identify what temporal patterns are present throughout the entire observation period, or put another way, to find out what temporal patterns are shared by all local segments of a sequence sample. Since these patterns exist continuously in the past, it is reasonable and straightforward to expect them to persist in the future.

Specifically, we introduce a within-sample contrastive learning mechanism to enable the encoder in $\mathcal{M}_{\text{fctst}}^{\text{us}}$ to extract persistent components from unstable patterns. Let $\{\mathbf{Z}_i^{\text{us}} \in \mathbb{R}^{D \times M \times H} | i = 1, 2, \dots, N\}$ denote the corresponding unstable representations of all samples. During training phase, apart from processing each $\mathbf{Z}_i^{\text{us}}$, the encoder in $\mathcal{M}_{\text{fctst}}^{\text{us}}$ also processes unstable representation of each subsequence, denoted as $\mathbf{R}_{i,m}^{\text{us}} \in \mathbb{R}^{D \times H}$, according to Eq. (12), where $m = 1, 2, \dots, M$. We denote the resulting encoded subsequence unstable representation by

$\hat{\mathbf{R}}_{i,m}^{\text{us}} \in \mathbb{R}^{D \times H}$. To learn patterns shared by all local periods, the post-encoding unstable representations of all subsequences from the same sequence sample should be guided to be similar with each other, while different from those from other sequence samples. Thus, we define a within-sample contrastive loss to achieve this:

$$\mathcal{L}_{\text{within}} = \frac{-1}{DNM(M-1)} \sum_{d=1}^D \sum_{i=1}^N \sum_{m=1}^M \sum_{\substack{m^+=1 \\ m^+ \neq m}}^M \log \frac{\exp(\phi_d(\hat{\mathbf{R}}_{i,m}^{\text{us}}) \cdot \phi_d(\hat{\mathbf{R}}_{i,m^+}^{\text{us}})/\tau)}{\exp(\phi_d(\hat{\mathbf{R}}_{i,m}^{\text{us}}) \cdot \phi_d(\hat{\mathbf{R}}_{i,m^+}^{\text{us}})/\tau) + \sum_{j \neq i}^N \exp(\phi_d(\hat{\mathbf{R}}_{i,m}^{\text{us}}) \cdot \phi_d(\hat{\mathbf{R}}_{j,m}^{\text{us}})/\tau)}, \quad (13)$$

where $\phi_d: \mathbb{R}^{D \times H} \rightarrow \mathbb{R}^H$ is a function to extract d th element along the variate dimension, and representations with indices m and m^+ form a positive pair. In practice, we can also lower computational complexity by replacing $\mathcal{L}_{\text{within}}$ in Eq. (13) with $\mathcal{L}_{\text{within}}^{\text{sim}}$:

$$\mathcal{L}_{\text{within}}^{\text{sim}} = \frac{-1}{DN(M-1)} \sum_{d=1}^D \sum_{i=1}^N \sum_{\substack{m^+=1 \\ m^+ \neq m}}^M \log \frac{\exp(\phi_d(\hat{\mathbf{R}}_{i,m}^{\text{us}}) \cdot \phi_d(\hat{\mathbf{R}}_{i,m^+}^{\text{us}})/\tau)}{\exp(\phi_d(\hat{\mathbf{R}}_{i,m}^{\text{us}}) \cdot \phi_d(\hat{\mathbf{R}}_{i,m^+}^{\text{us}})/\tau) + \sum_{j \neq i}^N \exp(\phi_d(\hat{\mathbf{R}}_{i,m}^{\text{us}}) \cdot \phi_d(\hat{\mathbf{R}}_{j,m}^{\text{us}})/\tau)}, \quad (14)$$

$$\bar{m} \sim \mathcal{U}(\{1, 2, \dots, M\}), \quad (15)$$

which means that we randomly select one subsequence to be the anchor each time according to a uniform distribution $\mathcal{U}(\{1, 2, \dots, M\})$ and calculate the loss following Eq. (13). Similar to the across-sample case, conditional on the current batch B , the following equations also hold for $\mathcal{L}_{\text{within}}^{\text{sim}}$:

$$\mathbb{E}_{\bar{m}}[\mathcal{L}_{\text{within}}^{\text{sim}} | B] = \mathcal{L}_{\text{within}}, \quad \mathbb{E}_{\bar{m}}[\nabla_{\theta} \mathcal{L}_{\text{within}}^{\text{sim}} | B] = \nabla_{\theta} \mathcal{L}_{\text{within}}. \quad (16)$$

Thus, random anchor selection introduces no systematic bias, while repeated uniform sampling ensures that each subsequence has the same probability of serving as the anchor across training iterations. The proof is provided in Section 2 of the supplementary material.

Algorithm 1 Training

Require: Input samples $S = \{S_i | S_i = (\mathbf{X}_i, \mathbf{Y}_i), i = 1, 2, \dots, N_{\text{tr}}\}$, where N_{tr} is the number of training samples

Ensure: Model parameters θ

- 1: **repeat:**
 - 2: **for** $\mathbf{X}_i$ in $\{\mathbf{X}_i | i = 1, 2, \dots, N_{\text{tr}}\}$ **do**
 - 3: $\mathbf{Z}_i^s, \mathbf{Z}_i^{\text{us}} = \mathcal{M}_{\text{disent}}(\mathbf{X}_i)$, disentangle $\mathbf{Z}_i^s$ and $\mathbf{Z}_i^{\text{us}}$ according to Eq. (2)–(4)
 - 4: $\hat{\mathbf{Y}}_i^s = \mathcal{M}_{\text{fctst}}^s(\mathbf{Z}_i^s)$, obtain $\hat{\mathbf{Y}}_i^s$ according to Eq. (12)
 - 5: $\hat{\mathbf{Y}}_i^{\text{us}} = \mathcal{M}_{\text{fctst}}^{\text{us}}(\mathbf{Z}_i^{\text{us}})$, obtain $\hat{\mathbf{Y}}_i^{\text{us}}$ according to Eq. (12)
 - 6: $\hat{\mathbf{Y}}_i = \mathcal{M}_{\text{int}}(\hat{\mathbf{Y}}_i^s, \hat{\mathbf{Y}}_i^{\text{us}})$
 - 7: **end for**
 - 8: calculate $\mathcal{L}$ according to Eq. (17)
 - 9: take gradient step on $\nabla_{\theta} \mathcal{L}$
 - 10: **until** converged
-

3.4. Learning scheme

Ultimately, COLF is trained to minimize loss function $\mathcal{L}$:

$$\mathcal{L} = \mathcal{L}_{\text{fctst}} + \mu \mathcal{L}_{\text{across}} + \nu \mathcal{L}_{\text{within}}, \quad (17)$$

where $\mathcal{L}_{\text{fctst}}$ is specified as mean squared error to measure the forecasting error, and μ and ν are two hyperparameters to control the contributions of $\mathcal{L}_{\text{across}}$ and $\mathcal{L}_{\text{within}}$. Also, in practical computation, $\mathcal{L}_{\text{across}}$ and $\mathcal{L}_{\text{within}}$ can be replaced by their simplified versions $\mathcal{L}_{\text{across}}^{\text{sim}}$

Table 1
Statistics of the six datasets.

Dataset	D-variate	Time steps	Frequency
Exchange	8	7588	1 day
ETTh1	7	17,420	1 h
ETTh2	7	17,420	1 h
Weather	30	8760	1 h
Caiso	11	43,816	1 h
Solar	137	52,560	10 min

and $\mathcal{L}_{\text{within}}^{\text{sim}}$, respectively. The training and forecasting algorithms of COLF are presented in Algorithms 1 and 2, respectively. In addition, a detailed theoretical and empirical analysis of the computational efficiency and scalability of COLF is provided in the Section 3 of the supplementary material.

Algorithm 2 Forecasting

Require: Past sequences $\{\mathbf{X}_i | i = 1, 2, \dots, N_f\}$, where N_f is the number of forecasting samples

Ensure: Forecasted sequences $\{\hat{\mathbf{Y}}_i | i = 1, 2, \dots, N_f\}$

```

1: for  $\mathbf{X}_i$  in  $\{\mathbf{X}_i | i = 1, 2, \dots, N\}$  do
2:    $\mathbf{Z}_i^s, \mathbf{Z}_i^{\text{us}} = \mathcal{M}_{\text{disent}}(\mathbf{X}_i)$ , disentangle  $\mathbf{Z}_i^s$  and  $\mathbf{Z}_i^{\text{us}}$  according to Eq. (2)–(4)
3:    $\hat{\mathbf{Y}}_i^s = \mathcal{M}_{\text{fcst}}^s(\mathbf{Z}_i^s)$ , obtain  $\hat{\mathbf{Y}}_i^s$  according to Eq. (12)
4:    $\hat{\mathbf{Y}}_i^{\text{us}} = \mathcal{M}_{\text{fcst}}^{\text{us}}(\mathbf{Z}_i^{\text{us}})$ , obtain  $\hat{\mathbf{Y}}_i^{\text{us}}$  according to Eq. (12)
5:    $\hat{\mathbf{Y}}_i = \mathcal{M}_{\text{int}}(\hat{\mathbf{Y}}_i^s, \hat{\mathbf{Y}}_i^{\text{us}})$ 
6: end for

```

4. Experiments

4.1. Setup

4.1.1. Datasets

Six commonly used real-world datasets are used to test the performance of COLF, which are

- Exchange [46], collecting the daily exchange rates of eight countries including Australia, Britain, Canada, Switzerland, China, Japan, New Zealand and Singapore from 1990 to 2016.
- ETTh1 and ETTh2 [47], containing load and oil temperature data of two electricity transformers at two stations, recorded hourly from 2016/07 to 2018/07.
- Weather,² containing hourly data for 30 meteorological indicators in 2023.
- Caiso,³ containing hourly actual load of eleven zones recorded by CAISO (California ISO) from 2019/01 to 2023/12.
- Solar,⁴ containing solar power data sampled every 10 min from 137 PV plants in Alabama State in the year of 2006.

All datasets are chronologically segmented into training, validation and test sets at a ratio of 6 : 2 : 2, and they are first preprocessed with Z-score standardization, where the mean and standard deviation are computed exclusively from the training set and subsequently used to normalize the validation and test sets, before the experiments. The statistics of the six datasets are presented in Table 1.

² <https://www.bgc-jena.mpg.de/wetter/>.

³ <http://www.energyonline.com/Data/>.

⁴ <http://www.nrel.gov/grid/solar-power-data.html>.

4.1.2. Baselines

We select ten models as baselines for comparison, which are detailed as follows:

- FreqCycle [48]: a time-frequency-based forecasting model integrating explicit periodic modeling and segmented frequency-domain learning.
- TimePro [49]: a Mamba-based forecasting model constructing variable- and time-aware hyper-states to capture variable relationships and salient temporal information.
- iTransformer [45]: a Transformer-based forecasting model applying attention and feed-forward network to variate tokens.
- Koopa [27]: a Transformer-based forecasting model tackling non-stationarity with modern Koopman theory.
- TSMixer [33]: an MLP-based forecasting model with mixing operations along both temporal and feature dimensions.
- PatchTST [26]: a Transformer-based forecasting model with sequence patching and channel independence.
- MICN [22]: a convolution-based forecasting model efficient in capturing both local patterns and global correlations.
- DLinear [32]: a forecasting model primarily relying on linear layers to perform forecasting.
- Non-stationary Transformer [9]: a Transformer-based forecasting model tackling non-stationarity with stationarization, but specifically equipped with *De-stationary Attention* to alleviate over-stationarization.
- FEDformer [25]: a Transformer-based forecasting model with seasonal-trend decomposition and feature learning from both time domain and frequency domain.

4.1.3. Implementation

The COLF is trained for a maximum of 100 epochs, with early stopping based on the total loss on the validation set and a patience of 10 epochs. The learning rate is fixed at 0.01, the batch size is set to 512, and the optimizer is specified as Adam [50]. Also, to ensure reproducibility, the random seed is fixed at 3407 throughout all experiments. For evaluation metrics, MAE (mean absolute error) and MSE (mean squared error) are used to measure the forecasting performance of all models. All experiments related to neural networks are conducted on NVIDIA A10 24G GPUs.

For hidden dimension H , and hyperparameters μ and ν in loss function $\mathcal{L}$ in Eq. (17), a grid search is conducted to find out the optimal combination, where H is selected from {16, 32, 64, 128}, and μ and ν are both selected from {1.0, 0.1, 0.01, 0.001, 0.0001}. Notably, we set the K in Eq. (6) to 1 and the temperature parameter τ in all contrastive losses to 1.0 during the grid search, and further determine their optimal values after acquiring the optimal (H, μ, ν) , by selecting K from {1, 2, 3, 4, 5} and τ from {0.01, 0.1, 1.0}. Besides, other hyperparameters are set to fixed values, with the subsequence length l set to 16, the stride s set to 8, the number of layers in all Transformer Encoders used set to 1, and the number of heads of all attentions used set to 8. Further discussion of the hyperparameters is provided in Section 4 of the supplementary material.

4.2. Main results

4.2.1. Quantitative comparison

The comparison results between the proposed COLF and all baselines are presented in Table 2. As shown, COLF achieves the best performance in around 70% of the cases. In particular, COLF consistently outperforms all baseline models across different forecasting horizons and evaluation metrics on datasets Exchange and Caiso. And even in cases where it does not achieve the best performance, COLF generally ranks second or third, with only two exceptions occurring on the ETTh2 and Weather datasets.

Table 2

Comparison between the proposed COLF and baselines in terms of MAE and MSE on all six datasets, with $T = 96$ and $L \in \{48, 96, 144, 192\}$. The best results are highlighted in bold, while the second-best results are marked with underlines.

Models	COLF		FreqCycle		TimePro		iTransformer		Koopa		TSMixer		PatchTST		MICN		DLinear		Nonstat.		FEDformer		
Metrics	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	
Exchange	48	0.1603	0.0531	0.2304	0.0998	0.1696	0.0595	0.1715	0.0600	0.1703	0.0594	0.2990	0.1706	0.1636	0.0579	0.2327	0.0964	0.1899	0.0679	0.1956	0.0762	0.2560	0.1219
	96	0.2249	0.1005	0.2823	0.1493	0.2382	0.1133	0.2358	0.1092	0.2364	0.1108	0.4505	0.3602	0.2316	0.1110	0.2566	0.1182	0.2568	0.1198	0.2767	0.1483	0.3092	0.1775
	144	0.2798	0.1516	0.3320	0.2031	0.2887	0.1625	0.2857	0.1570	0.2882	0.1611	0.5529	0.5485	0.2839	0.1600	0.3227	0.1791	0.3152	0.1778	0.3433	0.2262	0.3594	0.2354
	192	0.3261	0.2010	0.3761	0.2578	0.3382	0.2189	0.3303	0.2048	0.3280	0.2032	0.6482	0.7488	0.3375	0.2241	0.3806	0.2442	0.3674	0.2401	0.3907	0.2991	0.4016	0.2930
ETTh1	48	0.4186	0.3806	0.4837	0.4957	0.4121	0.3839	0.4203	0.3967	0.4323	0.4126	0.4559	0.4304	0.4126	0.3742	0.4375	0.4033	0.4266	0.4091	0.5135	0.5493	0.4550	0.4222
	96	0.4566	0.4350	0.5082	0.5307	0.4559	0.4477	0.4569	0.4511	0.4696	0.4701	0.5044	0.5057	0.4509	0.4252	0.4789	0.4583	0.4602	0.4597	0.5398	0.5869	0.4929	0.4793
	144	0.4758	0.4635	0.5262	0.5605	0.4724	0.4740	0.4802	0.4849	0.4919	0.5025	0.5397	0.5596	0.4798	0.4728	0.5030	0.4892	0.4809	0.4877	0.5921	0.7005	0.5287	0.5308
	192	0.4933	0.4922	0.5368	0.5779	0.4940	0.5098	0.4945	0.5060	0.5077	0.5273	0.5691	0.6081	0.5018	0.5112	0.5171	0.5090	0.4969	0.5092	0.5844	0.6788	0.5281	0.5340
ETTh2	48	0.2913	0.1860	0.3116	0.2055	0.2947	0.2066	0.2956	0.1978	0.2991	0.1965	0.3233	0.2118	0.2980	0.1987	0.3050	0.1967	0.2963	0.1905	0.3296	0.2236	0.3299	0.2308
	96	0.3198	0.2239	0.3368	0.2415	0.3172	0.2261	0.3227	0.2292	0.3293	0.2362	0.3927	0.2959	0.3202	0.2264	0.3330	0.2328	0.3346	0.2331	0.3860	0.2990	0.3489	0.2616
	144	0.3367	0.2457	0.3516	0.2628	0.3441	0.2592	0.3415	0.2509	0.3432	0.2514	0.5478	0.5560	0.3440	0.2530	0.3678	0.2682	0.3567	0.2576	0.3881	0.3044	0.3613	0.2804
	192	0.3526	0.2680	0.3624	0.2796	0.3559	0.2731	0.3495	0.2650	0.3537	0.2676	0.5608	0.5871	0.3499	0.2635	0.3820	0.2822	0.3697	0.2727	0.4007	0.3233	0.3733	0.2995
Weather	48	0.2416	0.1887	0.2881	0.2393	0.2467	0.1989	0.2498	0.1989	0.2629	0.2107	0.2797	0.1992	0.2418	0.1934	0.2938	0.2313	0.2679	0.2035	0.2948	0.2609	0.2904	0.2380
	96	0.2904	0.2491	0.3189	0.2835	0.2880	0.2501	0.2924	0.2528	0.3083	0.2730	0.3182	0.2446	0.2809	0.2431	0.3530	0.2778	0.3058	0.2514	0.3320	0.3253	0.3223	0.2791
	144	0.3177	0.2809	0.3438	0.3162	0.3244	0.2937	0.3215	0.2916	0.3387	0.3136	0.3322	0.2646	0.3136	0.2818	0.3494	0.2831	0.3328	0.2848	0.3546	0.3740	0.3478	0.3113
	192	0.3438	0.3154	0.3632	0.3387	0.3491	0.3226	0.3491	0.3213	0.3634	0.3425	0.3454	0.2832	0.3404	0.3108	0.4031	0.3326	0.3547	0.3088	0.3788	0.4081	0.3688	0.3344
Caiso	48	0.2268	0.1251	0.3156	0.2008	0.2309	0.1315	0.2360	0.1348	0.2821	0.1718	0.2735	0.1590	0.2459	0.1440	0.2508	0.1401	0.2811	0.1723	0.2560	0.1491	0.3070	0.1865
	96	0.2889	0.1860	0.3563	0.2541	0.3034	0.2053	0.2899	0.1910	0.3252	0.2211	0.3152	0.2045	0.2986	0.1985	0.3095	0.2052	0.3223	0.2193	0.3210	0.2124	0.3542	0.2450
	144	0.3191	0.2204	0.3673	0.2722	0.3523	0.2563	0.3207	0.2263	0.3464	0.2479	0.3361	0.2292	0.3220	0.2255	0.3285	0.2268	0.3414	0.2433	0.3529	0.2679	0.3746	0.2728
	192	0.3329	0.2358	0.3852	0.2942	0.3705	0.2797	0.3380	0.2456	0.3631	0.2695	0.3494	0.2443	0.3380	0.2445	0.3429	0.2412	0.3543	0.2593	0.3640	0.2707	0.3943	0.2947
Solar	48	0.1902	0.1449	0.3922	0.3691	0.2155	0.1806	0.2453	0.1938	0.3006	0.2449	0.2403	0.1513	0.2891	0.2222	0.2446	0.1657	0.4360	0.3181	0.2260	0.1921	0.4564	0.3721
	96	0.2171	0.1765	0.3407	0.3250	0.2324	0.1949	0.2498	0.2053	0.2987	0.2539	0.2697	0.1834	0.2745	0.2151	0.2260	0.1889	0.3743	0.2839	0.2397	0.2095	0.3579	0.2675
	144	0.2297	0.1963	0.3182	0.3041	0.2441	0.2067	0.2523	0.2089	0.3007	0.2604	0.2758	0.1874	0.2802	0.2322	0.2357	0.2018	0.3529	0.2715	0.2328	0.2050	0.3386	0.2462
	192	0.2446	0.2117	0.3595	0.3579	0.2598	0.2283	0.2728	0.2361	0.3222	0.2985	0.3000	0.2140	0.2931	0.2429	0.2504	0.2134	0.3929	0.3111	0.2660	0.2498	0.3618	0.2691
1st count		33		0		3		1		0		3		8		0		0		0		0	

Moreover, note that these baseline models generally incorporate internal mechanisms to tackle non-stationarity, either through normalization, seasonal-trend decomposition, or latent pattern disentanglement. Therefore, the superior performance of COLF demonstrated by these results further justifies the effectiveness of the proposed two contrastive learning mechanisms in tackling non-stationarity. For instance, TimePro and iTransformer rely on normalization to tackle non-stationarity, and COLF outperforms both in nearly all cases, with the few exceptions confined to several MAE comparisons with TimePro and the ETTh2 setting at a prediction horizon of 192 for iTransformer. FEDformer is a typical model with seasonal-trend decomposition, and COLF outperforms it across all scenarios, with an average reduction of 19% in MAE and 23% in MSE. Compared with Koopa, which applies latent pattern disentanglement based on Koopman theory, COLF still shows a significant advantage. The only exception is a slight degradation (from 0.2676 to 0.2680) in the MSE on dataset ETTh2 at a prediction horizon of 192, whereas COLF achieves better performance than Koopa in all other scenarios.

In summary, these comparison results are quite encouraging and serve as strong evidence of the powerful predictive capability of our model COLF.

4.2.2. Statistical test

To further examine whether the superior performance of COLF holds statistical significance, we conduct the Friedman test and post-hoc Bonferroni–Dunn test [51] on the performance rankings of COLF and four runner-up models, TimePro, iTransformer, TSMixer and PatchTST. Each model is ranked according to its relative performance in each (dataset, metric, L) group. For example, in group (Exchange, MAE, 48), COLF is ranked 1, TimePro is ranked 3, iTransformer is ranked 4, PatchTST is ranked 2, and TSMixer is ranked 5.

The test results are reported in Fig. 5. As displayed, the p -value of the Friedman test is much smaller than the significance level α , which means that the null hypothesis is rejected and these five models exhibit significant differences in performance. Moreover, the corresponding Kendall's W statistic is 0.4020, indicating a moderate overall

effect size and a meaningful degree of consistency in model rankings across the evaluated experimental settings. Based on this, the post-hoc Bonferroni–Dunn test results further indicate that COLF significantly outperforms the other four models as the ranking gap between COLF and each of the others exceeds the critical distance. In addition, as reported in Table 3, COLF's significant ranking advantage is also accompanied by practical improvements in forecasting accuracy, with positive average reductions in both MAE and MSE against all these four competing models, although the magnitude of the improvements varies across datasets.

4.2.3. Qualitative comparison

The previous two subsections quantitatively demonstrate the superior predictive performance of the proposed COLF. In this subsection, we further visualize and compare the forecasted results of COLF with those four runner-up baseline models PatchTST, iTransformer, TimePro and TSMixer on dataset Caiso, to provide a more intuitive illustration of its advantages. The visualized forecasted results are shown in Fig. 6. As displayed, it can be clearly observed that the forecasted curves of COLF fit the ground-truth curves most closely. On one hand, the forecasted curves of COLF exhibit evidently smaller deviations from the ground-truth curves compared with the other four models. On the other hand, the forecasted curves of COLF also show the best shape alignment with the ground-truth curves. All these results provide an intuitive complement to the quantitative analysis, further demonstrating the superiority of our proposed model, COLF.

4.3. Analysis

4.3.1. Ablation study

Recall that the proposed COLF tackles non-stationarity by disentangling temporal patterns into stable and unstable patterns, and conducts forecasting by integrating the prediction results derived separately from the two patterns. To further verify the effectiveness of this way of forecasting, we conduct an ablation study to evaluate the forecasting

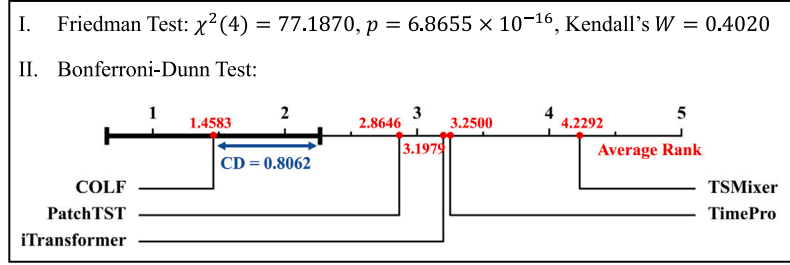


Fig. 5. Results of Friedman test and Bonferroni–Dunn test (significance level $\alpha = 0.05$). The Friedman test results are reported in the upper part, labeled as I, including the χ^2 statistic, Kendall's W statistic and p -value. The Bonferroni–Dunn test results are presented through a diagram in the lower part, labeled as II, which displays the average ranking of each model in red, the critical difference (CD) in blue, and the comparison intervals with thick black lines.

Table 3

Mean and standard deviation of the cross-dataset relative reductions in MAE and MSE achieved by COLF over PatchTST, iTransformer, TimePro and TSMixer.

	COLF vs. PatchTST	COLF vs. iTransformer	COLF vs. TimePro	COLF vs. TSMixer
MAE reduction (%)	4.62 $\pm$ 8.89	3.61 $\pm$ 5.07	3.33 $\pm$ 3.31	20.02 $\pm$ 15.86
MSE reduction (%)	6.20 $\pm$ 7.67	5.54 $\pm$ 4.31	6.75 $\pm$ 3.89	21.79 $\pm$ 28.17

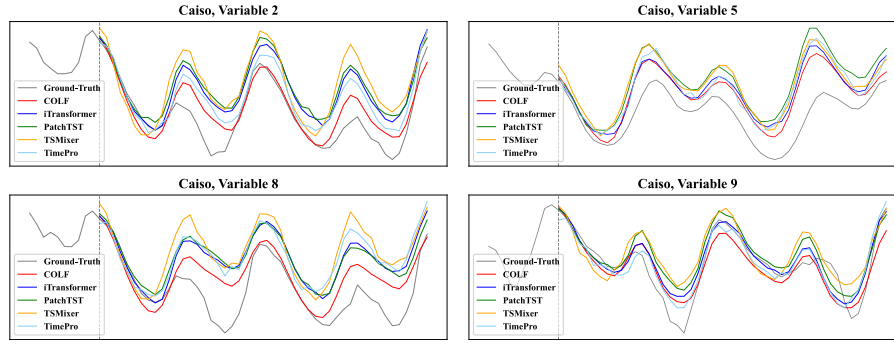


Fig. 6. Visualized forecasted results of COLF, PatchTST, iTransformer, TimePro, and TSMixer on dataset Caiso. Only the results for four variables (2, 5, 8 and 9) are shown, as the differences are most visually distinguishable in these cases.

Table 4

Comparison of COLF, COLF with only Z^s and COLF with only Z^{us} in terms of MAE and MSE on all six datasets, with $T = 96$ and $L \in \{48, 96, 144, 192\}$. The best results are highlighted in bold.

Datasets		Exchange		ETTh1		ETTh2		Weather		Caiso		Solar	
Metrics		MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
COLF	48	0.1603	0.0531	0.4186	0.3806	0.2913	0.1860	0.2416	0.1887	0.2268	0.1251	0.1902	0.1449
	96	0.2249	0.1005	0.4566	0.4350	0.3198	0.2239	0.2904	0.2491	0.2889	0.1860	0.2171	0.1765
	144	0.2798	0.1516	0.4758	0.4635	0.3367	0.2457	0.3177	0.2809	0.3191	0.2204	0.2297	0.1963
	192	0.3261	0.2010	0.4933	0.4922	0.3526	0.2680	0.3438	0.3154	0.3329	0.2358	0.2446	0.2117
Only Z^s	48	0.2173	0.0933	0.5471	0.6375	0.3244	0.2292	0.2946	0.2561	0.3152	0.1999	0.5011	0.3722
	96	0.2508	0.1170	0.5777	0.6467	0.3302	0.2396	0.3196	0.2947	0.3709	0.2638	0.3085	0.2268
	144	0.3093	0.1725	0.5620	0.6256	0.3512	0.2626	0.3389	0.3121	0.3831	0.2802	0.3168	0.2475
	192	0.3707	0.2535	0.5626	0.6253	0.3803	0.3049	0.3798	0.3667	0.3907	0.2903	0.3153	0.2609
Only Z^{us}	48	0.2042	0.0758	0.4886	0.4877	0.3324	0.2256	0.2420	0.1894	0.4872	0.3985	0.9877	1.3240
	96	0.2740	0.1430	0.5175	0.5447	0.3596	0.2694	0.2914	0.2492	0.4816	0.4011	0.6292	0.7207
	144	0.3252	0.1999	0.5483	0.5908	0.3555	0.2678	0.3188	0.2866	0.4908	0.4203	0.5797	0.6507
	192	0.3416	0.2152	0.5663	0.6302	0.3529	0.2687	0.3440	0.3156	0.5117	0.4539	0.6198	0.6684

performance of COLF when relying on either disentangled pattern alone. The results are shown in Table 4.

As illustrated, compared with the fully-structured COLF, neither COLF with only Z^s (forecasting based solely on stable patterns) nor COLF with only Z^{us} (forecasting based solely on unstable patterns) is able to deliver comparable prediction outcomes. The complete COLF achieves an average reduction of 18% in MAE and 23% in MSE relative to COLF with only Z^s , while achieving an average reduction of 24% in MAE and 31% in MSE compared with COLF with only Z^{us} .

Overall, these results validate the necessity and effectiveness of the pattern disentanglement and re-integration designed in COLF from an experimental perspective.

4.3.2. Pattern contributions under varying data non-stationarity

After verifying the advantages of jointly modeling the stable and unstable patterns in Section 4.3.1, we further investigate how these two patterns respectively contribute to forecasting under different levels of data non-stationarity.

Following the research paradigm, we use the Augmented Dickey–Fuller (ADF) statistic [52,53] to measure the non-stationarity of a time

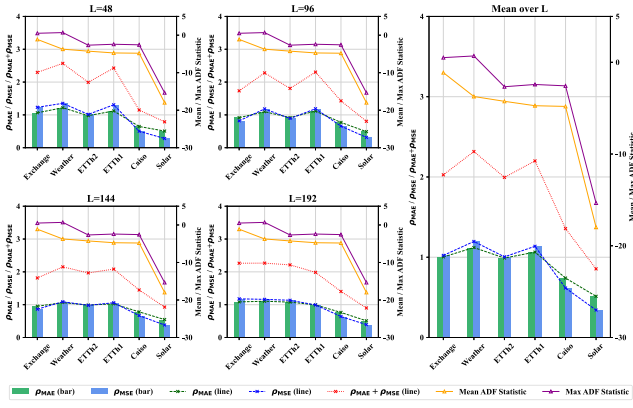


Fig. 7. Relationships between ρ_{MAE} and ρ_{MSE} , and the ADF statistics. The four plots on the left illustrate how ρ_{MAE} , ρ_{MSE} and $\rho_{MAE} + \rho_{MSE}$ vary with the ADF statistics under each prediction length L , while the large plot on the right shows how their averaged values across all prediction lengths change with the ADF statistics.

series. A larger value of the ADF statistic indicates a higher degree of non-stationarity. Since all data used in this article are multivariate time series, we calculate, for each dataset, the mean and maximum values of the ADF statistics over all its univariate component series. To quantify the contribution of each pattern to prediction, we introduce metrics ρ_{MAE} and ρ_{MSE} :

$$\rho_{MAE} = \frac{MAE_s}{MAE_{us}}, \quad \rho_{MSE} = \frac{MSE_s}{MSE_{us}}, \quad (18)$$

where MAE_s , MSE_s denote the MAE and MSE obtained by relying solely on stable patterns, and MAE_{us} and MSE_{us} denote the MAE and MSE obtained by relying solely on unstable patterns. Note that the contribution measured by these two metrics does not refer to an explicitly assigned predictive weight for either pattern, but rather to the relative contribution to predictive effectiveness characterized by the relative magnitudes of the standalone forecasting errors of the stable and unstable-pattern branches. For both ratio metrics, we primarily focus on their relative variations across datasets. If either ratio is higher for one dataset than for another, it indicates that unstable patterns make a greater relative contribution to prediction on the former dataset than on the latter, or equivalently, that stable patterns make a smaller relative contribution on the former dataset.

The relationships between ρ_{MAE} and ρ_{MSE} , and the ADF statistics are illustrated in Fig. 7. It can be clearly observed that metrics ρ_{MAE} , ρ_{MSE} and $\rho_{MAE} + \rho_{MSE}$ in all subplots show an overall downward trend as the data non-stationarity weakens. Although fluctuations appear at certain points in cases other than $L = 192$, the overall downward trend is highly pronounced. From the pattern-disentanglement perspective in this article, weaker non-stationarity generally implies a larger proportion of stable components and a smaller proportion of unstable components in a time series sample. Therefore, the greater relative predictive contribution of stable patterns, or equivalently, the lower relative predictive contribution of unstable patterns, observed as non-stationarity weakens across datasets, provides further empirical evidence that COLF can effectively distinguish and disentangle stable and unstable patterns in non-stationary time series.

4.3.3. Effect of K and τ

Recall that the two-level contrastive learning framework of COLF involves two key hyperparameters, K and τ . The K controls positive-sample coverage, with larger values reducing sampling variance at higher computational cost. The τ controls contrastive sharpness, where smaller values strengthen discrimination but increase sensitivity to difficult pairs, whereas excessively large values may weaken representation separation. In this subsection, we proceed to examine the impact

of these two hyperparameters on the overall predictive performance of COLF.

The Fig. 8 exhibits the sensitivity of COLF's performance to hyperparameters K and τ in terms of both MAE and MSE. For K , over 60% of the optimal K values are no less than 3, and within this subset, more than half are concentrated at 4 or 5. This indicates that a larger K can generally lead to better forecasting performance, which is an expected outcome since K serves as an approximation factor and a larger K means more accurate loss calculation. For τ , the proportions of optimal values at 0.01, 0.1, and 1.0 are approximately 12%, 40%, and 48%, respectively. Note that around 88% of the optimal τ values are found at either 1.0 or 0.1, with $\tau = 1.0$ alone accounting for nearly 50% of all cases. This reveals a clear preference of COLF for moderate to large τ values. Considering that τ controls the sharpness of contrastive discrimination, this preference may suggest that the stable and unstable representations produced by the disentangling attention are already separable to some extent, making an overly sharp contrastive constraint less necessary. Moreover, a relatively large τ may also help prevent positive representations from becoming overly homogeneous and reduce overfitting to incidental noise at both contrastive levels.

4.3.4. Representation collapse evaluation

In COLF, the two contrastive losses are jointly optimized with the forecasting MSE loss, which provides a task-oriented constraint against trivial or collapsed representations by requiring the learned representations to retain information useful for forecasting. As a complementary empirical validation, we calculate the average feature variance and effective rank [54] of the representations involved in the two contrastive learning mechanisms. Specifically, for across-sample contrastive learning, we evaluate the stable and unstable representations, Z^s and Z^{us} , respectively. For within-sample contrastive learning, we evaluate the encoded subsequence and the encoded full-sequence unstable representations, $\hat{R}^{us}$ and $\hat{Z}^{us}$, respectively. The results are reported in Table 5. As shown, for across-sample contrastive learning, the variances of Z^s and Z^{us} remain above 0 across all datasets, while their effective ranks range from 4.5621 to 29.3203 and from 5.6707 to 36.2932, respectively. This indicates that both representation branches preserve non-trivial variation along multiple latent directions, suggesting that the simplified across-sample objective does not lead to degenerate representations. For within-sample contrastive learning, $\hat{R}^{us}$ and $\hat{Z}^{us}$ also exhibit nonzero variances, with effective ranks ranging from 2.9978 to 13.0124 and from 2.1358 to 13.0941, respectively. Overall, these results provide consistent evidence against collapse in the representations learned by the two contrastive learning mechanisms.

4.3.5. Separability evaluation of disentangled representations

To evaluate the effectiveness of pattern disentanglement, we first quantitatively assess the separability between the learned stable and unstable representations. Specifically, we calculate the average cosine similarity between stable representations from different samples, denoted by $\text{Sim}(Z^s, Z^s)$, and that between stable and unstable representations, denoted by $\text{Sim}(Z^s, Z^{us})$. Also, we calculate the silhouette score [55] using cosine distance by treating the stable and unstable representations as two latent groups. The score ranges from -1 to 1 , with larger positive values indicating clearer separation, values near 0 indicating overlap, and negative values indicating possible group mismatch. All metrics are first calculated independently for each variate and then averaged across variates. As reported in Table 6, $\text{Sim}(Z^s, Z^s)$ is consistently higher than $\text{Sim}(Z^s, Z^{us})$ across all datasets, indicating that the stable representations exhibit stronger consistency across samples while remaining distinguishable from the unstable representations. The clear separability is also supported by the silhouette scores, which range from 0.4645 to 0.7585 across the six datasets and remain well above 0.

To provide a more intuitive illustration of the disentanglement effect, we visualize the disentangled stable and unstable representations

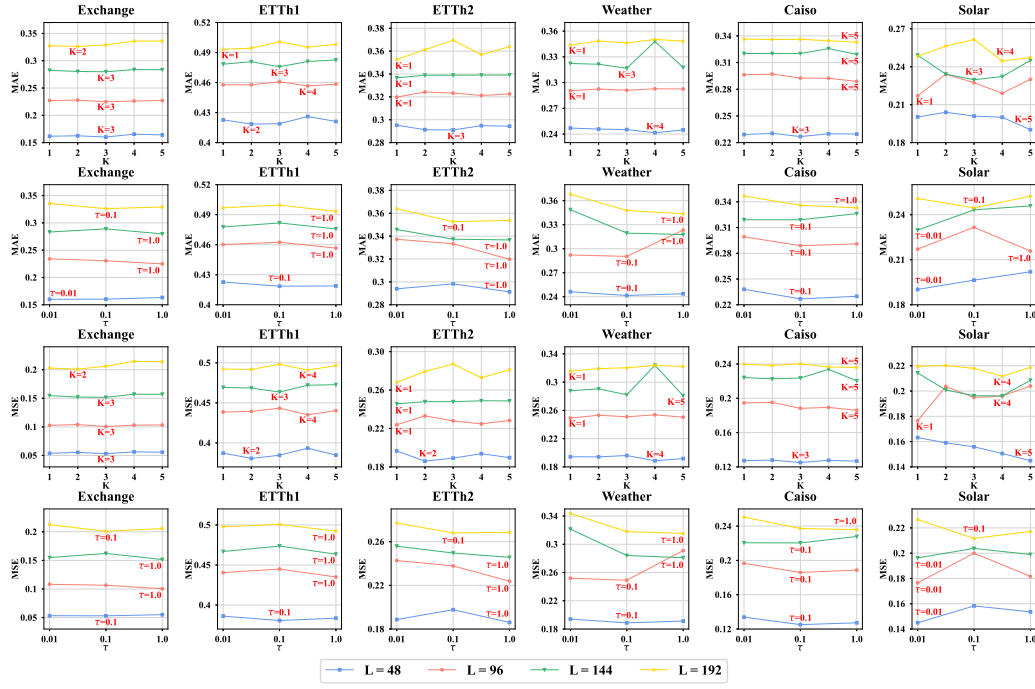


Fig. 8. Sensitivity of COLF's forecasting performance to hyperparameters K and τ on all six datasets in terms of both MAE and MSE. The optimal K or τ in each case is marked in red.

Table 5

Average feature variance and effective rank of representations involved in the across-sample and within-sample contrastive learning mechanisms.

Dataset	Across-sample contrastive learning				Within-sample contrastive learning			
	Z^s		Z^{us}		$\tilde{R}^{us}$		$\tilde{Z}^{us}$	
	Var.	Eff. rank	Var.	Eff. rank	Var.	Eff. rank	Var.	Eff. rank
Exchange	0.1819	11.6541	0.0805	10.5533	0.0052	2.9978	0.0935	2.1358
ETTh1	0.2565	29.3203	0.3375	25.9312	0.0048	9.6089	0.2085	7.9118
ETTh2	0.2342	19.5101	0.4288	22.5005	0.0603	5.6433	0.1449	4.4529
Weather	0.0082	4.5621	0.2744	5.6707	0.1553	10.8117	0.2197	9.2535
Caiso	0.0798	24.2208	0.0966	36.2932	0.0025	13.0124	0.1856	13.0941
Solar	0.0667	28.6107	0.0702	13.8585	0.0202	10.1176	0.0580	3.7000

Table 6

Values of $\text{Sim}(Z^s, Z^s)$, $\text{Sim}(Z^s, Z^{us})$ and the silhouette score on all six datasets.

Metrics	Exchange	ETTh1	ETTh2	Weather	Caiso	Solar
$\text{Sim}(Z^s, Z^s)$	0.8121	0.7265	0.7535	0.9920	0.6141	0.6329
$\text{Sim}(Z^s, Z^{us})$	0.3348	0.1029	-0.0229	0.4409	-0.1501	-0.1349
Silhouette	0.7585	0.4645	0.6452	0.6429	0.4749	0.6855

on datasets Exchange and ETTh2, as shown in Figs. 9(a) and 9(b), respectively. Since these representations are of high dimensionality, we only take their two-dimensional projections for visualization. As we can see, on both datasets Exchange and ETTh2, these two types of disentangled representations exhibit clear and visually intuitive separability. Especially on dataset Exchange, there is only a small overlapping region between the two groups. On dataset ETTh2, although the overlapping area is slightly larger, the overall separability is still remarkable.

Overall, both the quantitative metrics and visualizations validate the effectiveness of the proposed pattern disentanglement method.

5. Conclusion

In this paper, we propose a novel model, COLF, for non-stationary time series forecasting. COLF tackles non-stationarity through pattern disentanglement and performs forecasting by separately leveraging the disentangled patterns, which is primarily achieved through two

contrastive learning mechanisms operating at different levels. In this way, COLF is able to greatly alleviate the confounding effect caused by unstable patterns existing in non-stationary time series, and effectively exploit both stable and unstable patterns to facilitate more accurate and robust forecasting. We conduct extensive experiments and analyses to evaluate COLF from multiple perspectives, including quantitative comparison, ablation study, visualization and so on. The results demonstrate the overall superior performance of COLF over state-of-the-art approaches across different forecasting horizons and evaluation metrics, and further confirm the effectiveness of the internal design of COLF.

As a general method, COLF can benefit many application areas that rely on time series forecasting, such as financial investing, weather forecasting, energy forecasting and so on. However, in scenarios involving highly erratic or discontinuous temporal processes, the local smoothness assumption underlying the within-sample contrastive learning may not be fully applicable. In this regard, COLF could be further enhanced

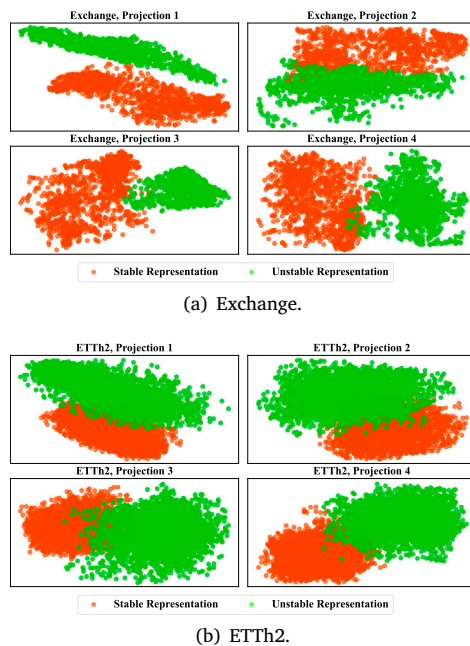


Fig. 9. Visualization of the disentangled stable and unstable representations on the Exchange and ETTh2 datasets.

by incorporating regime-aware mechanisms. Moreover, since the persistent components extracted from unstable patterns may not always persist over the forecasting horizon, estimating this uncertainty from the input context and adaptively adjusting their predictive contribution could further improve forecasting accuracy and robustness. Also, COLF could be extended with graph neural networks and causal modeling to explicitly capture multivariate correlations and accommodate multivariate causal interventions in real-world applications. In addition, COLF could also be combined with neural differential equations to support continuous-time forecasting, particularly in scenarios involving missing observations or irregular sampling intervals. We leave these topics for future investigation.

CRedit authorship contribution statement

Bowen Pang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Liyi Huang:** Writing – review & editing, Validation, Software, Resources, Methodology, Investigation, Data curation. **Xiangnan Feng:** Writing – review & editing, Validation, Investigation. **Wei Wei:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.patcog.2026.114808>.

Data availability

Data will be made available on request.

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